

The Equilibrium Constraint: One Killing Vector's Two Faces, the Geometric Lemma, and the Linearized Einstein Equation as an Equation of State

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****Abstract.**** The diamond paper formulated the entanglement-equilibrium problem in the model's own objects; this note executes it, with the claim level stated exactly. Three inputs are **proven**. First, a consistency lemma: the causal diamond's conformal Killing vector, restricted to the null boundary cone, equals $2\pi \lambda(\ell - \lambda)/\ell$ times the null tangent — machine-verified at 10^{-16} to be **precisely the series' interval weight**: the function modular theory forced on the generators and the function that weights Jacobson's matter integral are one vector field's two faces. Second, the matter side: the first-order first law $\delta S = \delta \langle K \rangle$ (rigorous because relative entropy is second order in the state perturbation), with the series' exact coherent law as the nonperturbative anchor pinning K 's normalization at 10^{-6} , and yielding separately the positivity $\Delta \langle K \rangle = S_{\text{rel}} \geq 0$ for entanglement-neutral excitations. Third, the geometric lemma: the area deficit of a small geodesic ball at fixed volume is $\delta A|_V = -\Omega_{d-1} \ell^{d+1} \hat{R}/2d(d+2)$, in $d=3$ equal to $-(4\pi/15)\ell^4 G_{00}$ by Gauss-Codazzi — derived, and verified numerically on explicitly curved metrics (two curvature classes, two radii, ratios 1.0000). Two hypotheses are then **postulated and named**: (E1) the total entropy variation carries a geometric term $\eta \delta A|_V$, the ultraviolet surrogate for gravitational degrees of freedom — a term whose algebraic rigorization is the crossed-product frontier, outside this note's claims; (E2) the vacuum is an equilibrium: the total first-order variation vanishes for all small diamonds in all frames at fixed volume. ****Theorem (conditional):**** given E1 and E2, the proven inputs force $G_{00} = (2\pi/\eta) \delta \langle T_{00} \rangle$ at every point in every frame — the two ball integrals conspiring to exactly 2π — and the all-frames polarization with the Bianchi identity yields

$$G_{ab} + \Lambda g_{ab} = \frac{2\pi}{\eta} \langle T_{ab} \rangle, \quad \eta = \frac{1}{4G} \Rightarrow 8\pi G,$$

with Λ a free integration constant. The linearized Einstein equation, as an equation of state — the metric a thermodynamic response variable, never an operator — closing the loop, at this conditional and linearized level, with the state-stratum reading the series' opening paper proposes for gravity.

1. The problem, as formulated

The diamond paper proved the exact first law on wedges and diamonds and stated the remaining demand: that $\delta S = \delta \langle K \rangle$ persist under *joint* metric and state variations for all small diamonds. This note extracts the content of that demand. Everything splits into what is proven, what is postulated, and what follows.

2. Lemma 1: one Killing vector, two faces

For the diamond with tips at $t = \pm\ell$ on the axis, the conformal Killing vector is $\zeta = \frac{2\pi}{2\ell} [(\ell^2 - t^2 - r^2) \partial_t - 2tr \partial_r]$. On the future null cone of the past tip ($t = \lambda - \ell, r = \lambda$), both components collapse to the same function:

$$\zeta|_{\text{cone}} = 2\pi \frac{\lambda(\ell - \lambda)}{\ell} (\partial_t + \partial_r) = 2\pi v_{(0,\ell)}(\lambda) k,$$

with $v_{(0,\ell)}$ exactly the interval weight of the diamond first-law paper (verified: residuals $8.9 \times 10^{-16}, 5.6 \times 10^{-17}$). On the central slice, $\zeta^t = 2\pi(\ell^2 - r^2)/2\ell$. The modular weight and the matter weight are one geometric object; the two papers' structures were never parallel constructions but two restrictions of one vector field.

3. The matter side, proven

First-order law. For a differentiable family of states ω_ϵ with $\omega_0 = \omega$, Araki's relative entropy satisfies $S(\omega_\epsilon || \omega) = O(\epsilon^2)$ — it is nonnegative and vanishes at $\epsilon = 0$ — so the identity $S_{\text{rel}} = \Delta \langle K \rangle - \Delta S$ gives, at first order, $\delta S = \delta \langle K \rangle$: the first law, rigorous, with K the diamond modular Hamiltonian of the companion papers. The series' exact coherent law is not this first-order statement but its nonperturbative anchor: it pins K 's geometry, weights, and constants at 10^{-6} , and it locates coherent excitations precisely: they carry second-order energy with $\Delta S = 0$ exactly, so they sit in the *second-order* sector of equilibrium, where the maximum condition is the inequality $\Delta \langle K \rangle = S_{\text{rel}} \geq 0$ — energy positivity for entanglement-neutral excitations — while the equation below comes from the first-order sector.

The ball integral. With $\delta \langle T_{00} \rangle$ approximately constant over a small ball, $\delta \langle K \rangle = 2\pi \int_{r < \ell} \frac{\ell^2 - r^2}{2\ell} \delta \langle T_{00} \rangle d^3x = \frac{8\pi^2}{15} \ell^4 \delta \langle T_{00} \rangle$.

4. Lemma 2: the geometric lemma, proven and verified

In Riemann normal coordinates on the spatial slice, $\sqrt{\det g} = 1 - \frac{1}{6} \hat{R}_{kl} x^k x^l + O(x^3)$, so for the geodesic ball of radius ℓ (radial lines being geodesics), $V = V_0(\ell) [1 - \frac{\hat{R} \ell^2}{6(d+2)}]$ and $A = dV/d\ell$. Comparing with the flat ball of *equal volume*:

$$\delta A|_V = - \frac{\Omega_{d-1} \ell^{d+1} \hat{R}}{2d(d+2)} \xrightarrow{d=3} - \frac{2\pi}{15} \ell^4 \hat{R} = - \frac{4\pi}{15} \ell^4 G_{00},$$

the last step by Gauss-Codazzi on a slice with vanishing extrinsic curvature at the center: $\hat{R} = 2G_{ab}u^a u^b$. Verified numerically on two explicitly curved metric classes — an isotropic space form ($\hat{R} = 6K$) and an $S^2 \times \mathbb{R}$ -type geometry ($\hat{R} = 2K$) — exploiting the Gauss-lemma fact that normal-form metrics make coordinate balls exactly geodesic, with the fixed-volume comparison solved exactly rather than linearized: ratios to the prediction 1.0000, 1.0000, 1.0000, 1.0001 at $\ell = 0.1, 0.2$.

5. The two hypotheses, named

(E1) The area term. The total entropy variation of a small diamond is postulated to be $\delta S_{\text{tot}} = \eta \delta A|_V + \delta S_{\text{matter}}$, with η a universal constant. The geometric term is the ultraviolet surrogate for gravitational degrees of freedom; on the series' type III₁ algebras entanglement entropy itself is undefined — only the relative entropy used throughout is finite — and the algebraic rigorization of ηA is exactly the generalized-entropy program of the crossed-product constructions (Witten and collaborators; cited as tradition, outside this note's claims). ***(E2) Equilibrium.*** The vacuum is stationary: $\delta S_{\text{tot}} = 0$ at first order, for all sufficiently small diamonds, in all frames, at fixed volume.

6. Theorem: the constraint and the equation

Theorem (conditional on E1, E2). For every point p and every unit timelike u :

$$G_{ab} + \Lambda g_{ab} = \frac{2\pi}{\eta} \langle T_{ab} \rangle.$$

$\eta \Big(\frac{4\pi}{15} \ell^4 G_{ab} u^a u^b \Big) + \frac{8\pi^2}{15} \ell^4 \langle T_{ab} \rangle u^a u^b = 0$
Longrightarrow $G_{ab} u^a u^b = \frac{2\pi}{\eta} \langle T_{ab} \rangle u^a u^b$, the two ball constants conspiring to exactly 2π ($\frac{8\pi^2}{15} \frac{4\pi}{15} = 2\pi$, displayed to six digits in the verification). Polarization over all unit timelike u gives $G_{ab} - \frac{2\pi}{\eta} \langle T_{ab} \rangle = \varphi g_{ab}$ for some function φ ; the contracted Bianchi identity and matter conservation force φ constant, $\varphi = -\Lambda$. Identifying the area coefficient with its Bekenstein-Hawking value, $\eta = 1/4G$, yields Newton's coupling exactly: $8\pi G$. The cosmological constant enters as an integration constant the argument leaves free — an output of honesty, not an oversight. $\blacksquare$

7. What closes, and what does not

At this conditional and linearized level, the loop to the series' opening paper closes: the metric has entered as a *state-stratum* quantity — a thermodynamic response variable in an equation of state, determined by entropy balance, never at any point an operator in the algebra — which is exactly the account of gravity's refusal of local observables that the revised Operator Mismatch paper proposes, now realized dynamically rather than merely located structurally. Every proven symbol in the balance is series-owned: the K 's and their verified weights, the Araki entropy on the III₁ algebras derived from the octonionic side, the stress tensor forced by the symplectic form, the reconstructed matter field. What does *not* close: E1 awaits its algebraic rigorization; the statement is linearized, with backreaction and nonlinear structure beyond it; the matter content is the model's own free field, and interactions — with the sector dynamics the soft-sector paper prescribed — remain ahead, together with the standing Steps 2 and 3 of the era.

8. Scope

Five items. (i) E1 and E2 are postulates, isolated by name; the theorem is conditional and claims nothing beyond the conditional. (ii) The geometric lemma is verified at leading

order with $O(K\ell^2)$ corrections visible in the numbers; the balance is a small- ℓ statement. (iii) The all-frames polarization and Bianchi step are elementary and included; matter conservation is the free field's. (iv) The measurement gate (Fewster-Verch on this net) remains the era's Step 2. (v) The covariance audit of the tower's finite- μ remnants remains Step 3 — and with the equation of state now on the table, that audit acquires its sharpest form: whatever interactions the tower natively supplies must respect the symmetry this balance presupposes.

References

- Companion papers: *The Diamond First Law: Interval Modular Hamiltonians, the Local Temperature Weight, and the Entanglement-Equilibrium Formulation; The First Law on the Net: The Stress Tensor, the Wedge Modular Hamiltonian, and the Relative Entropy of Coherent Excitations; The Interior Net: Local Algebras from the Cone and Microcausality as Output; The Assembly: Null Lines Through a Point, the Light-Cone Identification, and Lorentz Symmetry as Output.*
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Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.